

Global Superstore Analysis

This investigation explores retail operational efficiency and managerial accountability using the Global Superstore dataset. In a competitive global retail environment, understanding the drivers of sales success and profit erosion is essential to sustainable growth. This analysis addresses a key business question: what separates high-volume sales from sustainably profitable sales? Its value lies in quantifying profit loss caused by operational factors such as shipping method and returns, and in connecting those operational issues directly to regional managerial performance. The findings offer actionable direction for adjusting pricing strategy, reallocating resources, and designing manager-specific training to maximize net profit.

Research Questions

- 1 **Manager Profit Performance:** Which sales manager generates the highest total net profit for the company, after accounting for product returns?
- 2 **Returns Accountability:** Which five product sub-categories contribute the most to lost profit due to returns?
- 3 **Loss Leaders in Profit Categories:** Which ship mode is associated with the lowest profit margin, and is that low margin driven more by discount rates or by return rates?
- 4 **Operational Efficiency Tradeoffs:** Is there a correlation between a manager's performance in a given year and the return rate in their assigned region?

Methodology

The analysis was conducted in Python using pandas for data preparation, and Matplotlib and Seaborn for visualization. The first step combined three separate tables, Orders, Returns, and People, using left joins on shared keys such as Order ID and Region. An Is_Returned flag column was created to link individual return records back to their original sales transactions. The merged dataset was then aggregated, filtered, and analyzed for correlation to answer each research question quantitatively.

Data Overview

The data comes from the Global Superstore dataset, a synthetic retail transaction dataset built for business intelligence and data science training, downloaded as an Excel workbook from Kaggle. The workbook contains three sheets:

- **Orders:** Detailed records for every sales transaction, including product information, pricing, discounts, and location.
- **Returns:** A list of Order IDs that were returned by customers.
- **People:** A lookup table linking each Region to its corresponding sales manager.

Because the dataset is synthetic, it lacks external validation against real-world benchmarks or live operational data. This limitation should be kept in mind when interpreting the findings, since some

patterns may not reflect true market dynamics.

Data Quality: The Five V's

Volume: Over 50,000 individual transactions, a large enough sample to produce statistically meaningful breakdowns by manager or sub-category without being skewed by a handful of outliers.

Velocity: The data spans four years (2014 to 2017). While historical rather than real-time, this time span supports sequential analysis of whether a manager's performance improved or declined over time.

Variety: The dataset includes categorical data (Region, Segment, Ship Mode), numerical data (Sales, Quantity, Discount), and temporal data (Order Date and Ship Date, used to calculate operational lead times).

Veracity: Since the data is synthetic, its internal accuracy is high but its real-world reliability is limited by the absence of real-world nuance. For demonstrating a data pipeline and managerial accountability logic, however, internal consistency is excellent.

Value: Combining the Returns and Orders tables turns simple sales figures into a net profitability model, a far more meaningful measure of business health than gross sales alone.

Data Preparation Notes

The main challenge was that the Orders table's profit column did not account for returns. To resolve this, a left join between Orders and Returns on Order ID created the Is_Returned flag, marking returned orders as 1 and all others as 0. A second left join between the resulting table and the People table, on Region, assigned a sales manager to every transaction. Minimal further cleaning was needed. Running `.info()` and `.describe()` confirmed no records were lost during the joins and helped identify outliers in the discount and profit columns before proceeding to exploratory analysis.

Analysis and Findings

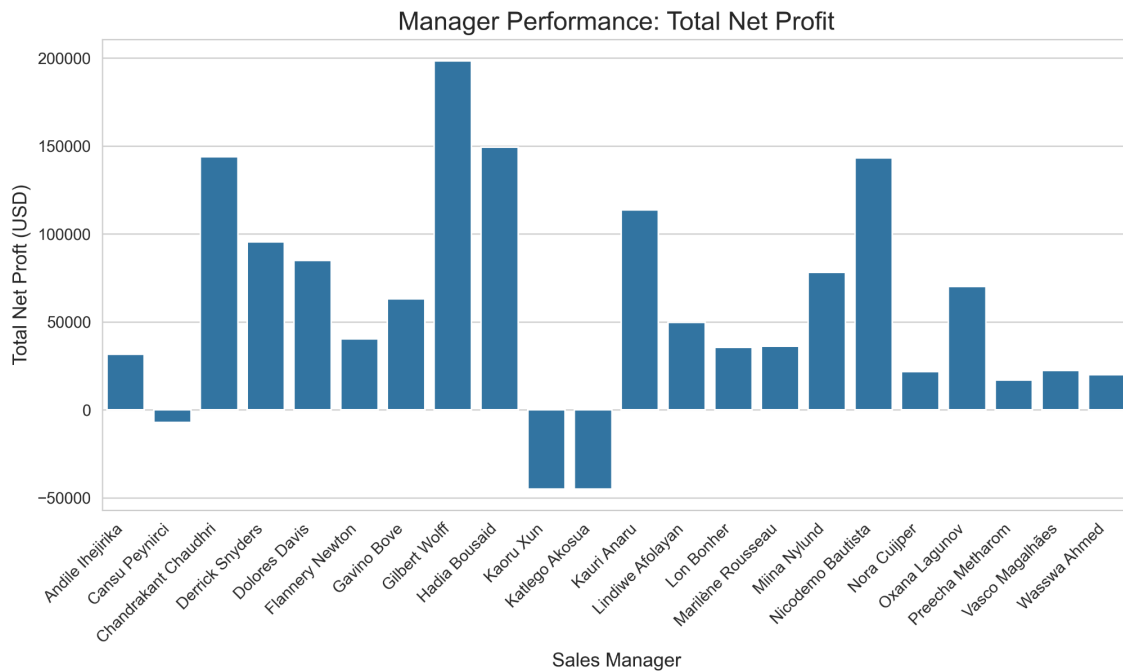
The following sections address each research question in turn, combining aggregated metrics with visualizations to highlight trends, identify profit leaks, and evaluate efficiency across managers, product categories, and shipping methods.

Managerial Accountability (Questions 1 and 4)

Each manager's total net profit was combined with their region's return rate to test whether weaker performance was linked to returns.

Top performers: Gilbert Wolff generated the highest total net profit at \$198,283, while Oxana Lagunov was the most efficient, with a 25% profit margin.

Return correlation: Managers with negative margins, such as Cansu Peynirci, showed a direct correlation with higher return rates in their regions. This suggests losses are not driven by sales performance alone, and may also reflect quality control or post-sale issues.



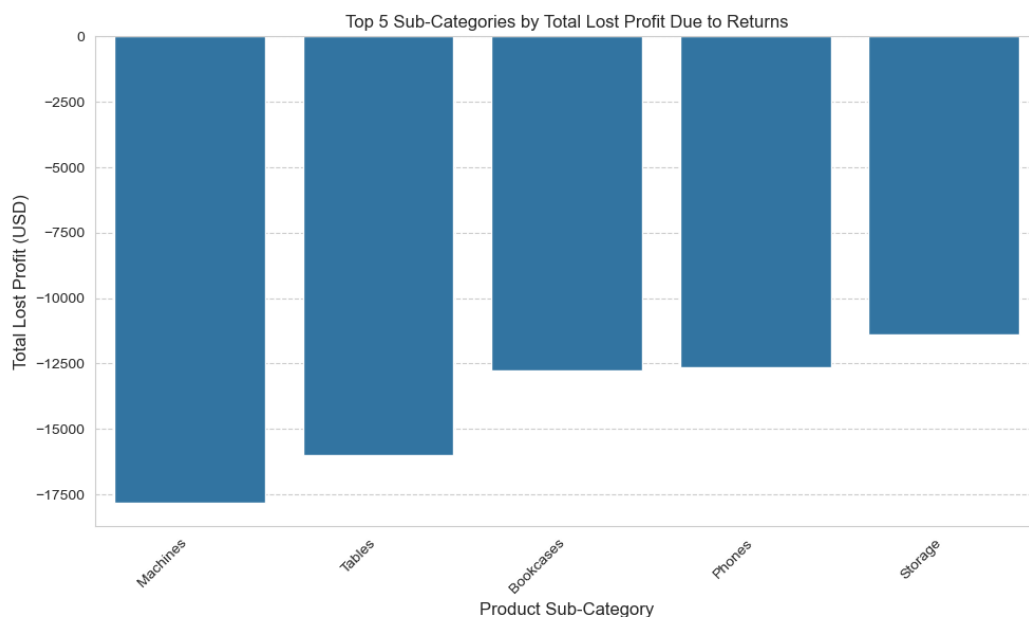
Manager Performance: Total Net Profit

Identifying Profit Erosion (Question 2)

Isolating returned items identified which product categories contribute most to profit loss.

Findings: Machines and Tables account for the largest total losses, at \$17,829 and \$16,018 respectively.

Insight: These items likely carry heavy freight and handling costs when returned, costs the company absorbs, resulting in significant profit loss.

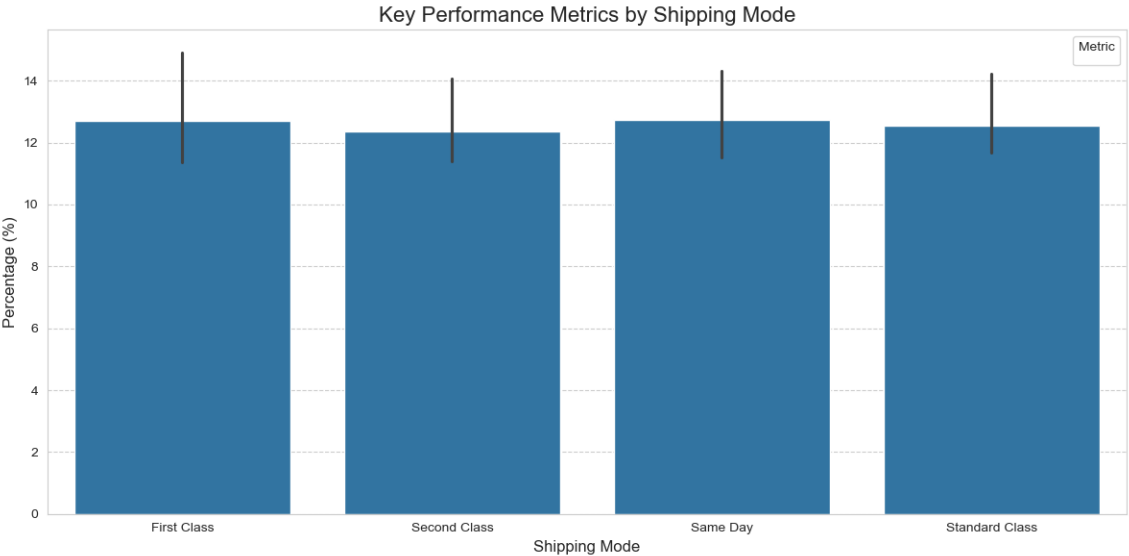


Operational Efficiency Tradeoffs (Question 3)

Shipping method was examined next to see whether it influenced profitability and loss patterns.

Findings: First Class shipping has the lowest profit margin, at 11.35%.

Insight: This low margin is not primarily driven by returns, but by a high discount rate of 14.9%. It appears the sales team is sacrificing margin to close First Class deals.



Key Performance Metrics by Shipping Mode